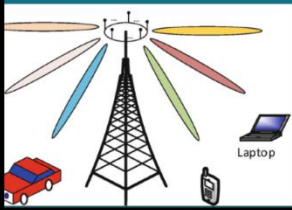


Experiment 10:



Differences between various multiple access techniques.

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Time Division Multiple Access (TDMA):

- **Concept:**
 - Divides the time into slots or frames.
 - Each user is assigned a specific time slot during which it can transmit.
- **Representation:**
 - Visualize time slots as distinct sections within a timeline.
- Label each section with the corresponding user or device.
- **Key Points:**
 - Users take turns to transmit during their designated time slots.
 - Efficient for digital communication, commonly used in GSM (Global System for Mobile Communications).

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Experiment 10:

Frequency Division Multiple Access (FDMA):

- **Concept:**
 - Allocates different frequency bands to individual users.
 - Each user is assigned a unique frequency channel for communication.
- **Representation:**
 - In a visual representation, you can use rectangles to represent different frequency bands.
- Label each rectangle with the corresponding user or communication channel.
 - **Key Points:**
 - Users transmit simultaneously, each using its allocated frequency band.
- Suitable for analog signals and FDMA is commonly used in analog radio communication. →

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Code Division Multiple Access (CDMA):

- **Concept:**
 - Allows multiple users to share the same frequency band simultaneously.
 - Each user is assigned a unique code to distinguish its signal from others.
- **Representation:**
 - Use triangles or other shapes to represent users, with unique patterns or colors representing their codes.
- Show overlapping areas to indicate simultaneous transmission.
 - **Key Points:**
 - Users transmit simultaneously on the same frequency using different codes.
- CDMA is widely used in digital cellular networks, such as in 3G and 4G technologies. →

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Experiment 10:

